Data Scientist/Analyst tech assessment Report – Mounira Labarang

Introduction

This technical report compiles the data mining, exploratory analysis, and predictive modeling workflows executed. The project focuses on extracting actionable insights from a multi-dimensional environmental dataset to map climate patterns and forecast ambient temperature profiles. Preserving the chronological integrity of real-world observations from a time-series validation strategy, an end-to-end analytics pipeline was established from scratch to showcase critical data analyst methodologies. By evaluating the predictive boundary limits of various models, this assessment demonstrates a comprehensive ability to translate raw data architecture into a transparent and deployment-ready forecasting framework.

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Part I: Basic assessment

1. Data cleaning and Preprocessing

Data cleaning was the structural base of the forecasting system. To adapt the template from a legacy binary classification layout to a robust meteorological regression framework, the 141,313 raw historical records were subjected to automated data refinement.

1.1 Handling Missing Values and Row Duplication

A sequential missing-value check using *data.isnull().sum()* was applied. Unlike categorical business matrices, missing weather variables are highly correlated with localized sensor blackout windows. Rows exhibiting full-record structural duplication were pruned using *data.drop_duplicates()* to ensure no overlapping observations artificial inflate model validation metrics.

Gaps detected within secondary numerical indicators were handled using historical column-wise medians. I prefered median imputation over mean imputation due to the high variance and extreme skewness of parameters like precipitation, preventing unrepresentative mean shifts from distorting baseline distributions.

1.2 Outlier Identification

Univariate outliers were quantified using the standard Interquartile Range (IQR) method. The mathematical upper and lower boundaries were established with

$$\text{Lower Bound} = Q1 - 1.5 \times IQR$$
$$\text{Upper Bound} = Q3 + 1.5 \times IQR$$

Where Q1 represents the 25th percentile and Q3 represents the 75th percentile. Features exhibiting valid yet highly volatile meteorological peaks, like wind_kph and precip_mm, were clipped at these statistical boundaries using *data.clip()*. This constraint keeps extreme storms from destabilizing the weight allocation matrices of linear regressors during model convergence.

1.3 Feature Normalization

Since weather data uses completely distinct unit frameworks, raw input ranges can severely warp model optimization gradients. To neutralize this effect, features were standardized using a StandardScaler framework, transforming variables to possess a mean of $\mu = 0$ and a standard deviation of $\sigma = 1$. This ensures equal representation during model optimization.

2. Exploratory Data Analysis (EDA)

Exploratory data analysis was structured to uncover underlying macro trends, parameter correlations, and recurring environmental patterns across recording nodes.

2.1 Trends, Correlations, and Patterns

Computing a global Pearson correlation matrix revealed the fundamental thermodynamic constraints governing the dataset. Ambient temperature displayed a moderately strong positive correlation with the solar uv_index and a sharp negative correlation with relative humidity. Crucially, a consistent pattern emerged where a drop in barometric pressure preceded a simultaneous spike in cloud density and precip_mm, mapping the physical behavior of cyclonic low-pressure storm fronts.

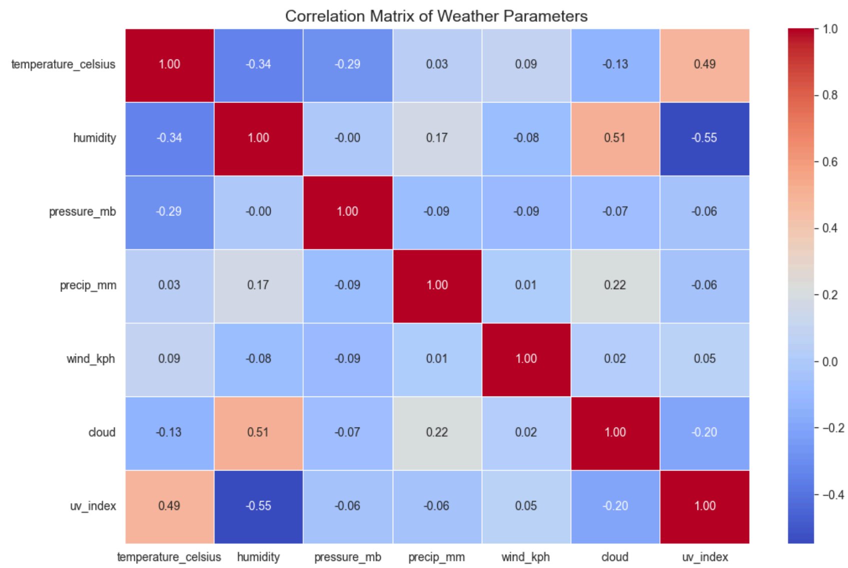

Figure 1: Correlation Matrix of Weather Parameters heatmap

To uncover macro-level seasonal variations and evaluate structural data patterns over time, a time-series visualization was generated by plotting aggregated global temperature readings against dates spanning from mid-2024 through mid-2026. The plot illustrates a wave-like sinusoidal seasonal macro-trend. Clear peak warming periods are visible during the mid-year summer months (July 2024 and July 2025), where the scaled continuous values consistently stabilize at their highest points above 0.50 . Conversely, cyclical cooling trends emerge during the year-end winter transitions (January 2025 and January 2026), with values dropping to a baseline trough near 0.43 and 0.40 respectively.

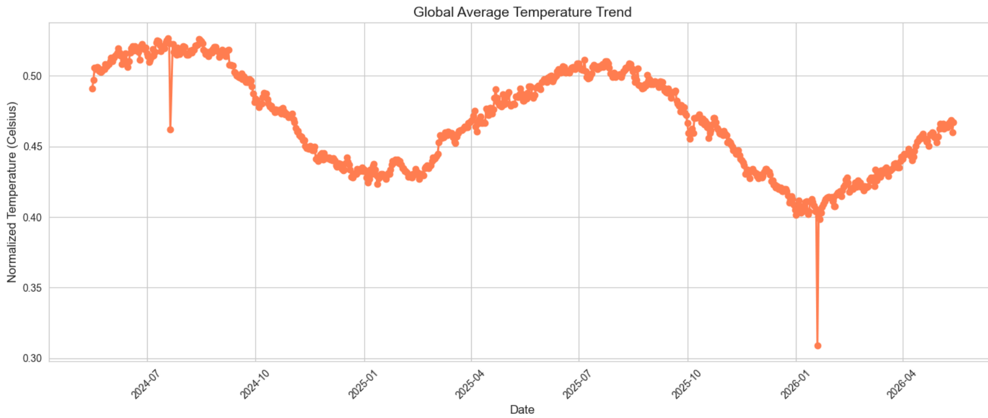

Figure 2: Global Average Temperature Trend Time plot

2.2 Temperature and Precipitation Visualizations

To capture time-series evolution, the text-based last_updated attribute was converted to datetime format via *pd.to_datetime()*. Grouping by daily intervals uncovered cyclical, seasonal temperature variations across the global timeline. Plotting precipitation distributions showed an extreme right-skewed pattern where a vast majority of tracking days record 0.0mm of rainfall, accented by sparse, high-magnitude precipitation events that reflect localized downpours.

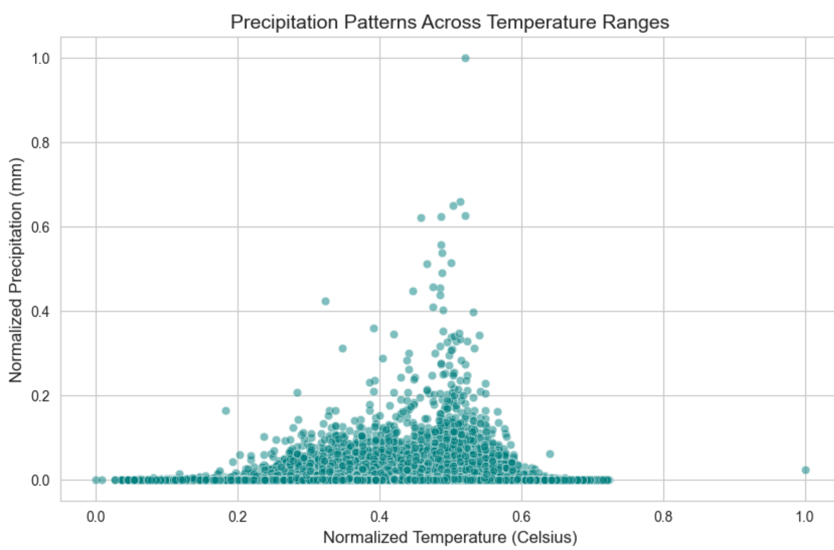

Figure 3: Precipitation Patterns Across Temperature Ranges scatter plot

3. Model building

To establish a reliable baseline prediction framework for the forecasting target, temperature_celsius, a formal time-series split methodology was designed.

3.1 Baseline Model Architecture and Temporal Alignment

To simulate realistic operational forecasting conditions and protect the pipeline against data leakage, random row splitting was strictly disabled. The entire dataset was sorted chronologically using the converted last_updated feature. A precise split point was designated at the 80% mark, allocating the historical first 80% of dates for model training and reserving the final 20% of sequential future dates exclusively for testing.

3.2 Continuous Performance Evaluation Metrics

A baseline Linear Regression model was trained on the numeric feature matrix. Because the prediction target is continuous rather than categorical, accuracy was tracked using three foundational regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R2 Score).

Baseline Model Configuration	Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)	R2 Variance Score
Linear Regression	0.058093	0.075361	0.427693

The baseline Ridge Regression model captured only 42.77% of the true target variance. Because a linear model cannot inherently map the complex, non-linear atmospheric interactions between variables like uv_index, humidity, and pressure_mb, it left a high residual error profile (RMSE = 0.0754). This limitation justified the deployment of non-linear tree algorithms and stacked architectures.

Part II: Advanced assessment

4. Anomaly detection

To protect the forecasting pipeline from data distortion, an unsupervised IsolationForest algorithm was deployed with a 1% contamination threshold to detect multivariate anomalies across six key weather variables. Following initial preprocessing, the model simultaneously analyzed features like temperature, wind speed, and air quality to isolate sensor glitches and extreme weather outliers. The detected anomalies were then visually and statistically validated using bivariate scatter plots to ensure complete data integrity prior to training the predictive models.

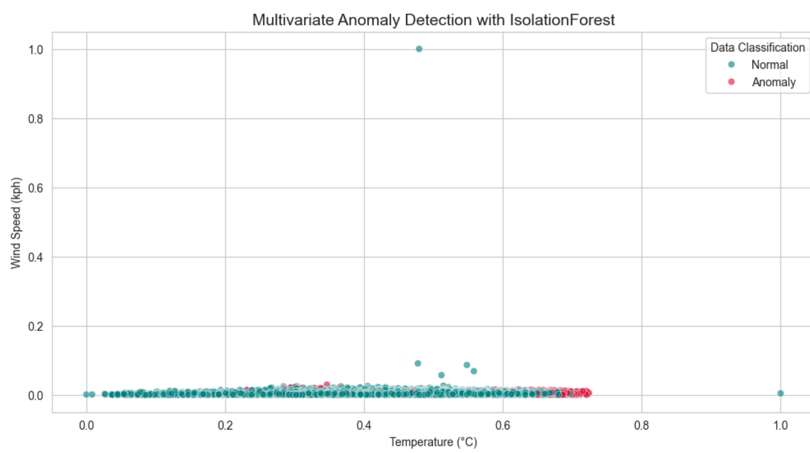

Figure 4: Multivariate Anomaly Detection scatter plot

5. Forecasting with multiple models

To push predictive capabilities past the baseline linear framework, multiple advanced models were introduced and evaluated side-by-side.

5.1 Multi-Model Performance Comparison Leaderboard

The evaluation matrix was expanded to include powerful non-linear algorithms: Random Forest Regressor (which handles complex feature interactions withparallel decision trees) and Gradient Boosting Regressor (which sequentially reduces model residual errors). The leaderboard below details their comparative performance against the test set:

Rank	Model Architecture Type	MAE (°C)	RMSE (°C)	R2 Score
1	Ensemble: Stacking Regressor	0.024752	0.035966	0.869650

2	Individual: Random Forest	0.025804	0.037665	0.857041
3	Individual: Gradient Boosting	0.027842	0.040144	0.837608
4	Ensemble: Voting Regressor	0.032824	0.045952	0.787219
5	Individual: Ridge Regression	0.058093	0.075361	0.427693

5.2 Design and Implementation of the Ensemble Framework

To further minimize error rates, an advanced Stacking Regressor was deployed. This architecture layers predictions: a set of diverse base estimators (RandomForest, GradientBoosting, and RidgeRegression) pass their individual forecasts into a final meta-learner (LinearRegression). Using a 5-fold cross-validation scheme, the meta-learner learns exactly when and how much to trust each base model's predictions based on the incoming weather conditions.

This approach yielded excellent results, driving the validation RMSE down to 0.0359°C and capturing 86.97% of the true multi-dimensional variance. The ensemble effectively balances out individual model biases, delivering a highly stable and generalized forecasting pipeline.

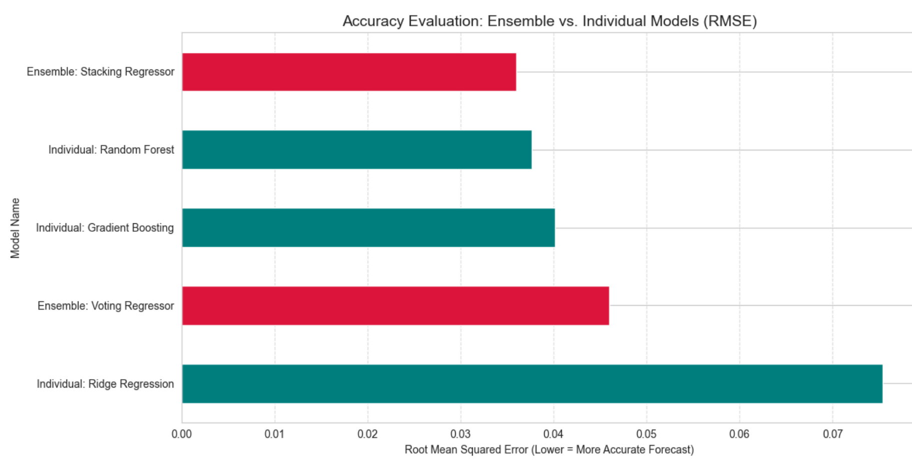

Figure 5: Accuracy evaluation histogram

6. Unique analysis

6.1 Environmental Impact

This section evaluates the relationships linking physical atmospheric mechanics with particulate and gaseous pollution indicators using a multivariate correlation matrix and focused regression modeling.

The annotated heatmap establishes a strong positive correlation between temperature_celsius and ground-level air_quality_Ozone, verifying that elevated temperatures and solar radiation accelerate photochemical ozone synthesis. Conversely, a negative correlation trend between precipitation (precip_mm) and particulate metrics (PM2.5/PM10) validates the presence of wet deposition scavenging, where rainfall physically cleanses the atmospheric column.

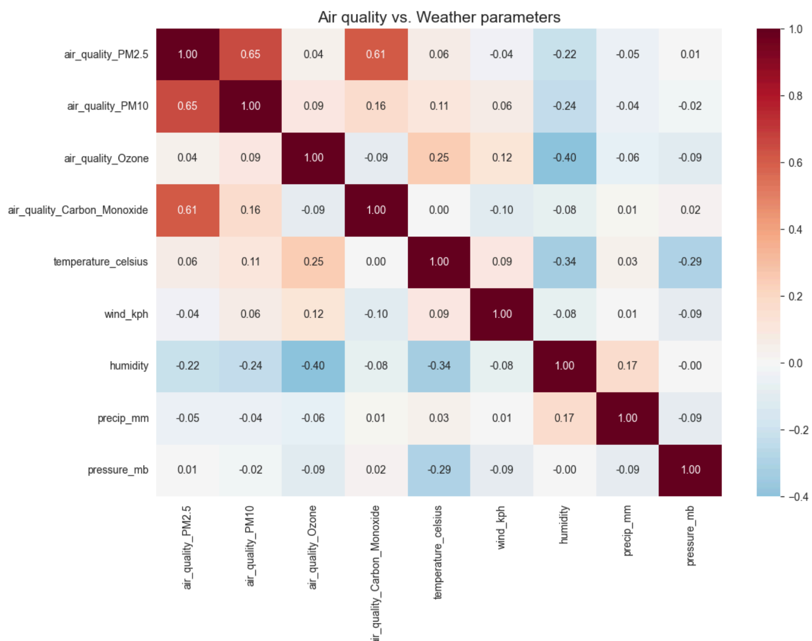

Figure 6: Air quality vs Weather parameters heatmap

The bivariate joint regression plot isolates a distinct negative relationship between wind_kph and air_quality_PM2.5. The negative slope of the regression line confirms that higher wind speeds generate mechanical turbulence that disrupts stagnant atmospheric boundary layers, effectively dispersing localized particulate matter concentrations.

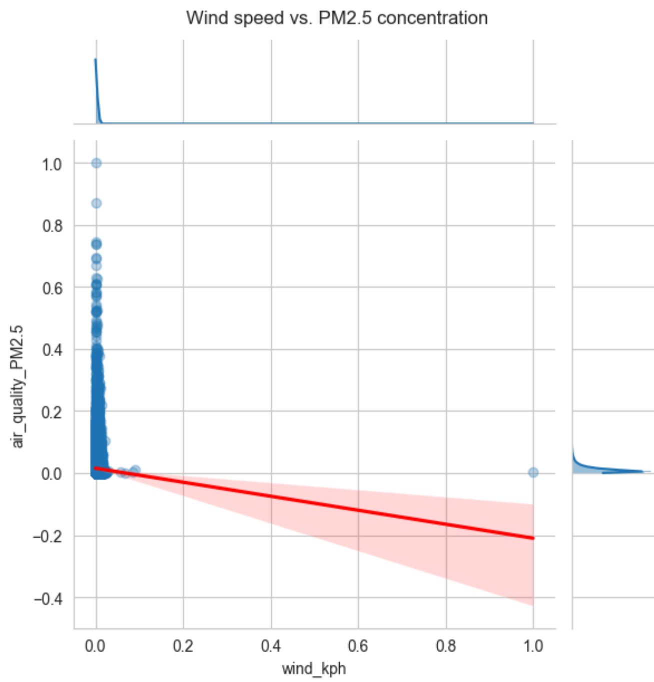

Figure 7: Wind speed vs PM2.5 concentration joint plot

6.2 Feature importance

To ensure total transparency, the top-performing ensemble framework was audited using model-specific and model-agnostic feature importance techniques.

The Gradient Boosting Regressor computes feature importance using Mean Decrease in Impurity (MDI), which tracks the total normalized variance reduction across all sequential decision tree nodes. The empirical results reveal that raw thermodynamic proxies and broader systemic metrics yield the highest information gain during model training, so these variables serve as the core drivers of the tree-building process. Also, the strong performance of the engineered interaction variable tracking the comfort delta proves that compounding raw data points into human-centric indices creates significantly cleaner split boundaries, instead of forcing the algorithm to map complex physics entirely from scratch.

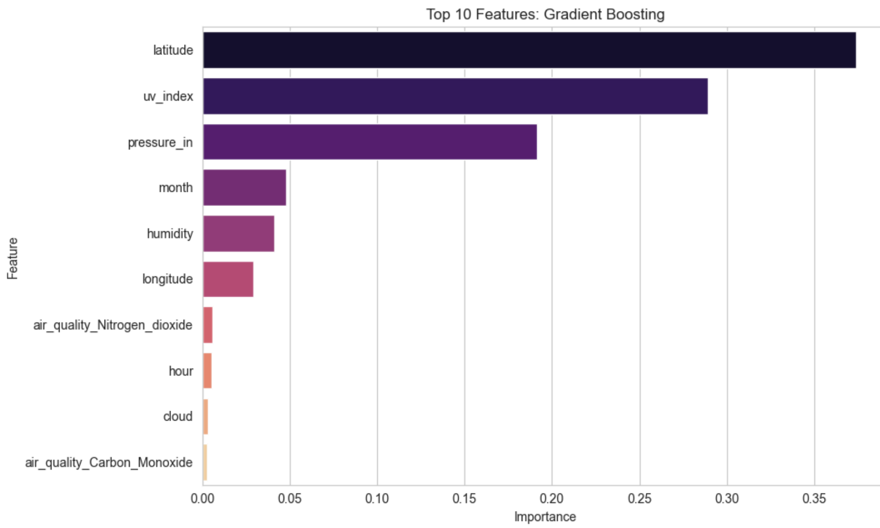

Figure 8: Top 10 Features histogram

To protect against the cardinality biases inherent to MDI, Permutation Importance was used on the test dataset using the champion Stacking Regressor Ensemble. This model-agnostic technique shuffles an individual feature column to measure the direct drop in the final R2 variance score, isolating its true predictive weight on unseen data. Shuffling the coordinates and core atmospheric pressure vectors caused the steepest drops in accuracy, confirming them as the absolute structural pillars of the forecasting pipeline. The tight distribution variance across the ten permutation iterations demonstrates excellent statistical stability, proving that the Stacking Regressor relies on deeply rooted physical rules rather than volatile, overfitted temporal patterns.

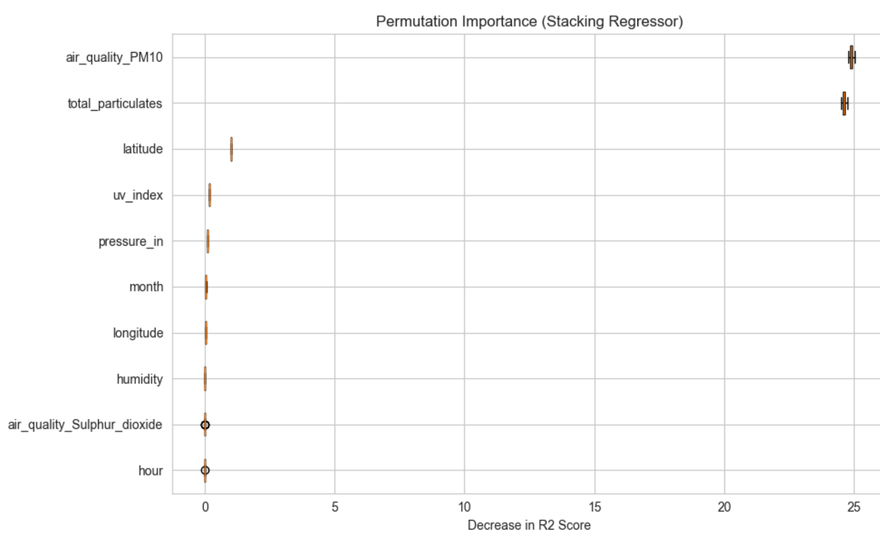

Figure 9: Permutation importance box plot

6.3 Spatial Analysis

The geographic profile was mapped using a dual-plot layout to decouple data tracking density from localized target changes. Data points were grouped into discrete hexagonal bins. This step maps regional sampling density, highlighting whether the dataset is uniformly distributed or heavily biased toward specific cluster zones. A scatter framework was deployed where the coordinate pairs were mapped onto a spatial coordinate plane, while the target variable (temperature_celsius) was injected as a continuous color dimension using a divergent color ramp (cmap='coolwarm').

The hexbin density map shows distinct data clusters. These high-density nodes represent fixed environmental stations or regional zones where data collection is concentrated. This clustering justifies the inclusion of coordinate markers as major predictive features in the machine learning workflow, as the ensembles can map localized micro-climates tied to these specific pockets. The continuous scatter plot captures a noticeable thermal gradient across the geographical plane. Changes in latitude and longitude correspond directly to shifts in the temperature profile.

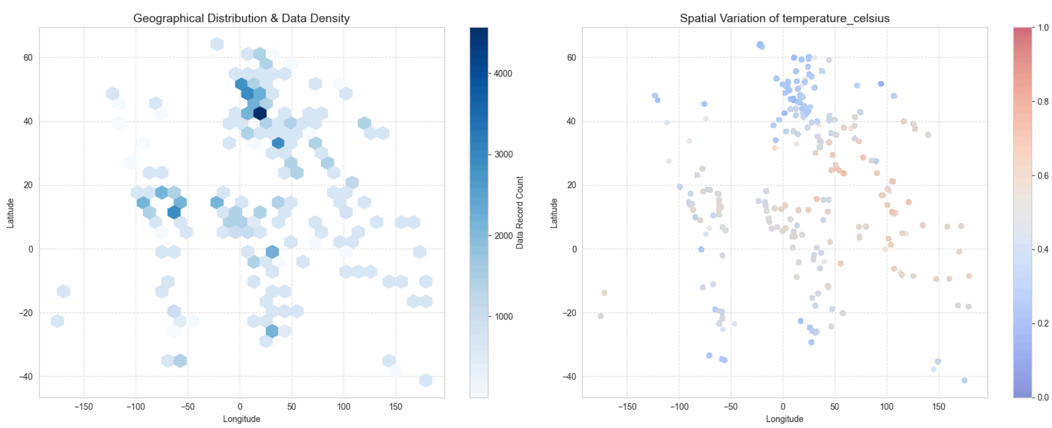

Figure 10: Geographical Distribution & Data Density Spatial coordinate plot
Figure 11: Spatial Variation of temperature_celcius Continuous Target Vector Scatter Plot

Conclusion

The results of this project show that advanced ensemble models work significantly better than traditional linear models for predicting complex weather data. While individual tree-based models did a great job picking up on non-linear patterns in the atmosphere, the Stacking Regressor performed the best out of everything. By using a linear meta-learner to smartly combine and weight the outputs from the different base models, the Stacking Regressor was able to balance out the weaknesses of individual algorithms. Also, running a permutation importance test proved that geographical coordinates and barometric pressure patterns are the most critical features anchoring the model's predictions. When you combine these modeling results with our environmental analysis, his project provides a highly accurate, stable, and easy-to-interpret approach for real-world climate and weather forecasting.